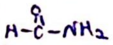
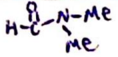
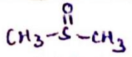
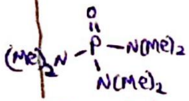
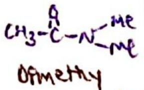
(Pre-Requisites for this chapter)

Solvent :

- Polar
  - Protic
    - → Presence of Proton
    - $\boxed{En-H}$ (Electronegative atom directly connected to -H)
  - Aprotic
    - → Absence of proton
    - $\boxed{En-G}$ (Electronegative atom connected to group/atom other than -H)
- Non-Polar
  - * Alkane
  - * Benzene
  - * CCl_4, etc

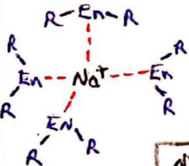
Polar Protic Solvents (PPS)	Polar Aprotic Solvents (PAS)
H_2O, R-OH, R-COOH,  $H-\overset{O}{\overset{\|}{C}}-NH_2$ (Formamide)	 $H-\overset{O}{\overset{\|}{C}}-N(Me)-Me$ Dimethyl Formamide (DMF),  $CH_3-\overset{O}{\overset{\|}{S}}-CH_3$ Dimethyl Sulphoxide (DMSO)  $(Me)_2N-\overset{O}{\overset{\|}{P}}(-N(Me)_2)-N(Me)_2$ Hexamethyl Phosphonamide (HMPA)  $CH_3-\overset{O}{\overset{\|}{C}}-N(Me)-Me$ Dimethy Acetamide (DMA)

Nucleophilicity :

→ It is affinity of an attacking Reagent towards +ve charge density.

When Solvent is PAS :→

(NaX salts are dissolved in the solvent)

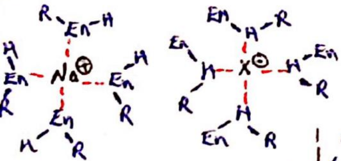


The counter ion Na^+ is solvated by solvent, so salts wont recombine in the solvent.

Stability : $I^{\ominus} > Br^{\ominus} > Cl^{\ominus} > F^{\ominus}$

$\boxed{\text{Nucleophilicity}: F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}}$

When Solvent is PPS :→



Here, both cation & anion are solvated.

→ So here, high charge density (size↓, EN↑) then high solvation & low Nucleophilicity

Solvation : $F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

$\boxed{\text{Nucleophilicity}: I^{\ominus} > Br^{\ominus} > Cl^{\ominus} > F^{\ominus}}$

Eg: Which is better $Nu^{\ominus}$ in PPS :

$OH^{\ominus} < SH^{\ominus}$ ($OH^{\ominus}$ will be solvated more)

General Trend :→

Basicity, $Nu^{\ominus}$ ↓ses →

Along the periods

Down the group ↓ Basicity ↓ses, $Nu^{\ominus}$↑ (PPS), $Nu^{\ominus}$↓ (PAS)

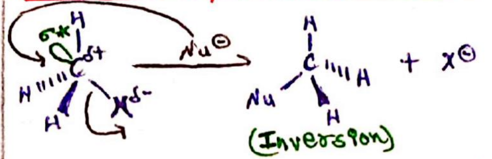
NOTE: If solvent not specified, then assume PPS by default.

Haloalkanes & Haloarenes -1

(*) Types of Nucleophilic Substitution :

1) S_N1
2) S_N2
3) S_Ni
4) S_N1'
5) S_N2'
6) S_Ni'
7) $S_N NGP$
8) $S_N AE$
9) S_N2AE
10) S_N1Ar

Bimolecular Nucleophilic Substitution (S_N2) :



(Inversion)

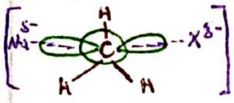
(*) Walden Inversion takes place for product

(*) Bimolecular Rx^n $\boxed{Rate = K[CH_3X][Nu^{\ominus}]}$

(*) One step Reaction

Imp (*) R & d may or may not convert to S & l.

(*) sp^2 Transition state (TS)



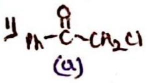
(*) Highly unstable TS due to $10e^-$ on C-atom

↳ Implies, EWG would increase rate of rx^n Imp

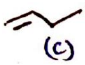
Factors Affecting S_N2 :→

1) Conc. of $Nu^{\ominus}$ must be high for good yield
2) Nucleophilicity must be good. R.O.R.
3) Hinderance must be less $[1°X > 2°X > 3°X]$
4) Leaving group X must be good. $[R-I > R-Br > R-Cl > RF]$
5) EWG would increase Rate of Rx^n.

Ques compare rate of SN^2 for following $Nu^{\ominus}$:

1)  $Ph-\overset{O}{\overset{\|}{C}}-CH_2Cl$ (a) CH_3Cl (b) CH_3CH_2Cl (c) $(CH_3)_3C-Cl$ (d)

Due to EWG (-Ph) → a > b (1°) Less bulky > c (1°) More bulky > d (3°)

2) $Ph-CH_2-Cl$ (a) CH_3-CH_2-Cl (b)  (c)

EWG → a > c > b (1°)

c: Resonance disperses e⁻ density & stabilises TS.

3) [NaOH] 1M [NaOH] 2M [NaSH] 2M

c > b > a ← Less conc. of $OH^{\ominus}$

c: Less solvation $Nu^{\ominus}$↑

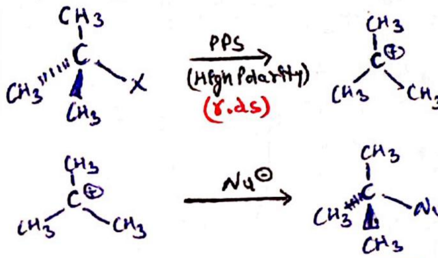
b: More conc. of $OH^{\ominus}$

Unimolecular Nucleophilic Substitution (S_N1)

→ Here, the C-X bond is broken by PPS.

→ We take PPS with high dielectric const/polarity to break C-X bond.

Eg of such PPS : H_2O, tBuOH, ROH, RCOOH



[Independent of conc. of $Nu^{\ominus}$]

(*) Unimolecular Rx^n $\boxed{Rate = K[RX]}$

(*) $\boxed{\text{Rate of } Rx^n \propto \text{Ease of dissociation of C-X bond}}$

↳ Actual Concept (RI > RBr > RCl > R-F)

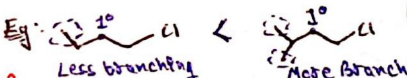
↳ $\boxed{\text{Rate of } Rx^n \propto \text{stability of Carbocation}}$

(Derived Concept.) (3°RX > 2°RX > 1°RX)

Imp (*) Thus, whenever degree of R-X is same, we check branching. Because

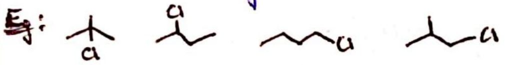
More Branching ⇒ More repulsion to -X (leaving group) ⇒ Easy C-X dissociation

i.e $\boxed{\text{Rate of } Rx^n \text{ when same degre RX} \propto \text{Branching}}$



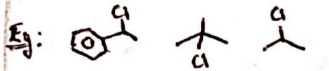
Less branching < More Branching (Rate of SN^1)

Imp (*) For R.O.R, we check stability of carbocation formed just after the C-X dissociation and not the Rearranged one.



Rate of S_N1 : 1 > 2 > 4 > 3

4 ← More Branching

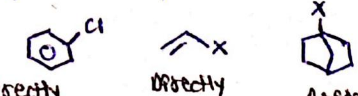


Rate of S_N1 ⇒ (1 > 2 > 3)

1 Due to Reso.

2° > 3° here

(*) Following species don't show any of the nucleophilic substitution (S_N1 & S_N2)



Directly attached to benzene | Directly attached to double bond | Bride head position

Reactions Shown by Haloalkanes :

(i) Ag has high affinity for 'X' (Cl, Br, I) and form ppts.

[AgCl → White ppt
AgBr → Pale Yellow ppt
AgI → Yellow ppt]

(ii) Zn, and Phosphorus (P) also have high affinity for oxygen.

$R-X \xrightarrow{AgF} RF + AgX\downarrow$

$R-X \xrightarrow{KCN} R-CN$ (Ionic)

$R-X \xrightarrow{AgCN} R-NC$ (Covalent)

Swarts Rx^n

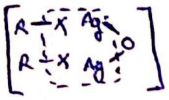
Can also be carried out with CaF_2, SbF_3, Hg_2F_2

(Ionic) $\xrightarrow{KNO_2}$ R-ONO

(Covalent) $\xrightarrow{AgNO_2}$ $R-NO_2$

$\xrightarrow[Ag_2O]{Moist}$ R-OH + AgX↓

$\xrightarrow[Ag_2O]{Dry}$ R-O-R



(S_N2) $\xrightarrow{NaOH}$ R-OH + NaX

(S_N1) $\xrightarrow{H_2O}$ R-OH + HX

$\xrightarrow{NH_3}$ $R-NH_2$ + HX

$\xrightarrow{R'COOAg}$ R'-COOR

$\xrightarrow{LiAlH_4}$ R-H

$\xrightarrow[(R')^{\ominus}(M)^{+}]{R'-M}$ R-R' + MX

Frenklestein / Halogen Exchange Reaction :→

$$R-X + NaI \xrightarrow[Acetone]{Dry} \underset{(S_N2)}{RI} + NaX\downarrow$$

→ For this Rxⁿ, precipitation of NaX is driving force (acc. to Le Chatlier's principle)

→ NaI is soluble in dry acetone because it is covalent while other NaX are insoluble in Acetone bcz they are ionic

→ Hence, w/o Dry acetone, this Rxⁿ can't take place.

Method of Preparation of Haloalkanes :

1) Rxⁿ of Alcohol with HX type Acid :→

$$R-OH + HX \longrightarrow R-X$$

Mechanism

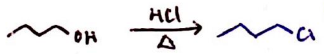
$$R-OH \overset{H^+}{\rightleftharpoons} R-\overset{\oplus}{O}H_2 \overset{(-H_2O)}{\rightleftharpoons} R^{\oplus} \overset{X^{\ominus}}{\rightleftharpoons} R-X$$

(*) Carbocation Intermediate

(*) Rearrangement Possible

(*) EDG would increase the rate.

Eg:



(when Δ given, then SN takes place instead of elimination)

HI : → No condⁿ req. for dissociation

↳ NaI/H_3PO_4 (✓)

NaI/H_2SO_4 (X) ← Because, H_2SO_4 oxidises NaI into I_2 instead of HI

HBr : → Slight heating req. for dissociation of HBr.

↳ $NaBr/H_3PO_4$ or H_2SO_4, both can be taken

HCl : → Heating req. for dissociation of HCl

↳ OR, take (dry HCl + Lewis Acid)

Example : Groove's Process / Luca's Test ⇒

R-OH $\xrightarrow{3°-OH}$ 3°-Cl (S_N1) ⇒ Immediate Turbidity

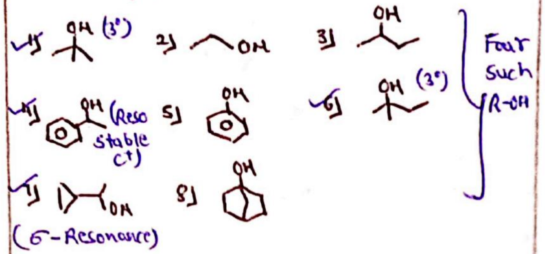
R-OH $\xrightarrow{2°-OH}$ 2°-Cl (S_N1) ⇒ Turbidity in 5-7 min

R-OH $\xrightarrow{1°-OH}$ 1°-Cl (S_N2) ⇒ No Turbidity at Room temp.

Reagent Used : Conc. HCl / Anhyd. $ZnCl_2$

↳ (Luca's Reagent)

Ques Which alcohols will show immediate turbidity ?



2) Rxⁿ of R-OH with Red P / Br₂ :→

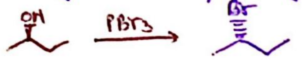
$$3\,R-OH \xrightarrow[(PBr_3)]{Red\,P/Br_2} 3\,R-Br + H_3PO_3$$

(*) S_N2 Mechanism

(*) Inversion

→ 3 moles of alcohol required for rxⁿ.

Eg:



3) Rxⁿ of R-OH with PCl₃ & PCl₅ :→

R-OH $\xrightarrow{PX_3}$ RX + H_3PO_3

R-OH $\xrightarrow{PX_5}$ RX + HX + POX_3

(*) Both have S_N2 Mechanism

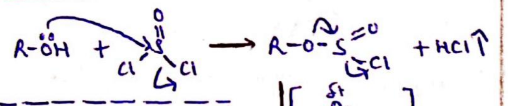
(*) Both have formation of P=O bond as driving force.

(*) Both gives inverted product.

4) Darzen's Process :→

$$R-OH + \underset{\text{(Thionyl chloride)}}{SOCl_2} \longrightarrow RCl + \underbrace{HCl\uparrow + SO_2\uparrow}_{\text{Gaseous Product}}$$

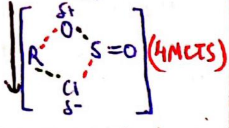
Mechanism ⇒



(*) 4-MCTS observed

(*) Intramolecular SN (S_Ni)

(*) Retention observed

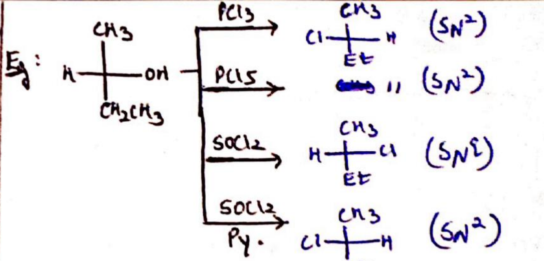


R-Cl + SO_2↑

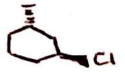
(*) Best M.O.P due to high yield because of formation of gaseous by products.

NOTE : If Bases like Pyridine or NR_3 are used with $SOCl_2$, then instead of S_Ni, S_N2 takes place.

Eg:

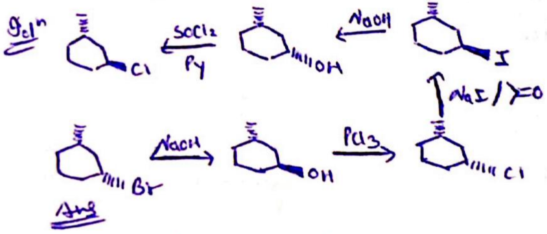


Eg. R-Br $\xrightarrow{\text{(i) NaOH (ii) PCl}_3\text{ (iii) NaI/acetone (iv) NaOH (v) SOCl}_2\text{/Py}}$



Find R-Br.

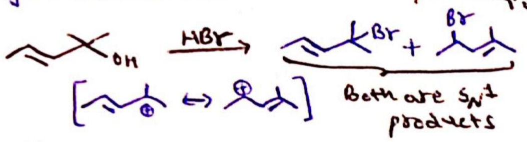
Solⁿ



S_N1' & S_N2' Reactions :

S_N1' Reaction :→ Rxⁿ in which product is obtained from the carbocation formed after Resonance.

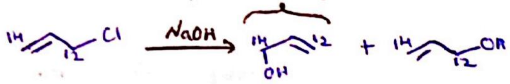
This one is also S_N1' product



Both are S_N1 products

S_N2' Reaction :→

Also S_N2' product



S_N2 products

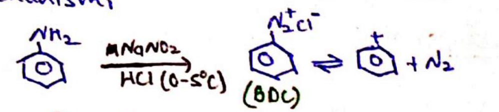
Haloarenes

M.O.P → (S_N1 Ar)

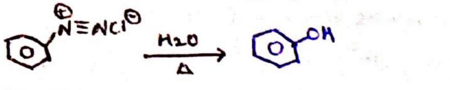
Chemical Rxⁿs → (S_N2 Ar), (EAS)

Method of Preparation of Haloarenes :

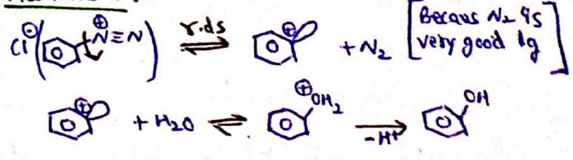
→ Reactions of Benzene Diazonium Chloride (BDC) occurs through S_N1-Ar mechanism.



Eg:



Mechanism :→



[Becaus N_2 is very good lg]

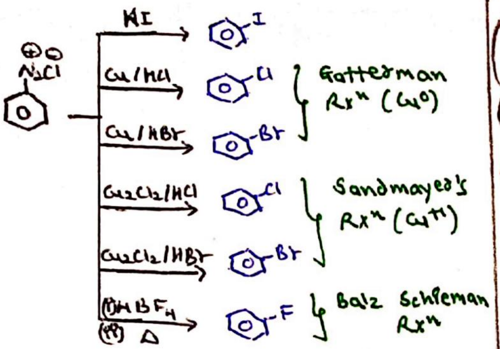
(*) Aryl cation intermediate

(*) Formation of cation is r.ds

(*) R.O.R is independent of Nu⊖

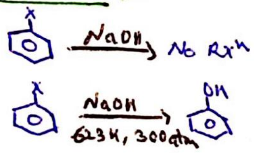
Haloalkanes & Haloarenes - 2

→ All following substitutions take place by S_N1-Ar mechanism.

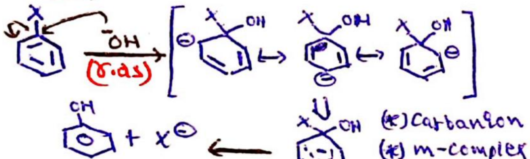


Chemical Rxn of Haloarenes:

Dow's Process:-



Mechanism =>



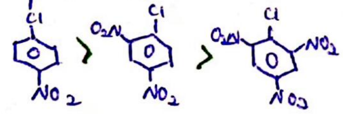
(*) Carbanion
(*) m-complex
(*) Meisenheimer Complex

→ -ve charge is dispersed on ortho/para positions in the intermediate.

→ Thus, EWG at ortho/para ⇒ Stabilises m-complex ⇒ R.O.R ↑ses

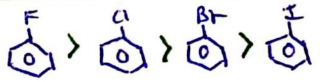
→ In case of normal temp. & pressure, S_N2-Ar would take place only if EWG present at ortho and para positions.

Rate of Rxn:

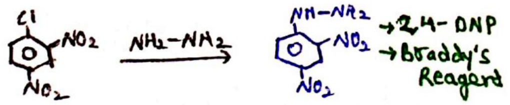


→ Also, $\boxed{\text{ROR} \propto -I \text{ effect of lg}}$

Rate of Rxn:

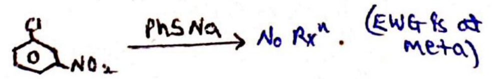


1)



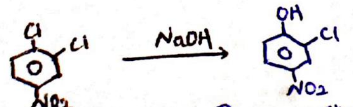
→ 2,4-DNP
→ Braddy's Reagent

2)

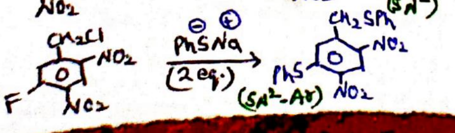


(EWG is at meta)

3)



4)

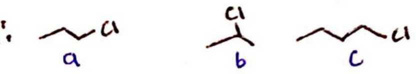


NOTE: S_N2 is faster than S_N2-Ar because it is single step rxn while the latter is two step rxn.

Boiling Point:

$$\boxed{\begin{aligned} \text{BP} &\propto \text{Molecular Weight} \\ &\propto \frac{1}{\text{Branching}} \end{aligned}}$$

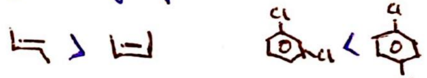
Eg:

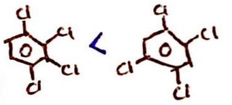


BP: c > a > b

Melting Point:

Melting Point of symmetric compounds is higher due to high packing efficiency of solids.





Basically jinke dipole vectors cancel ho rahe ho, unka MP ↑

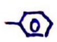
S_N-NGP (Neighbouring Group Participation):

→ NGP ↑ses rate of rxn.

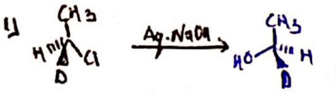
→ For NGP, compound must have
- ↳ internal Nu⊖
- ↳ lg & Nu⊖ anti to each other.

→ Atoms & groups which can show NGP ⇒

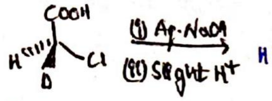
$-O^\ominus$	−NHR	$-C(=O)OR$	−I
$-NH^\ominus$	$-NR_2$	$-C(=O)NH_2$	−Cl
$-COO^\ominus$	−SR		−Br
$-NH_2$	−SH	$-CH=CH_2$	



1)

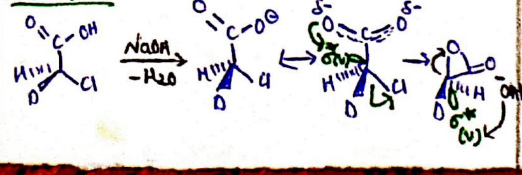


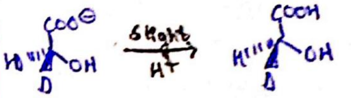
(S_N2)
(100% inverted product)



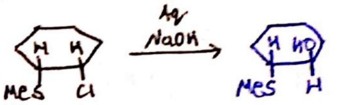
(SN-NGP)
(100% Retained product)

Mechanism

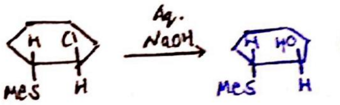




2).

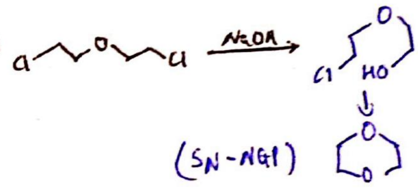


(S_N2)



(S_N-NGP)

3)



(S_N-NGP)

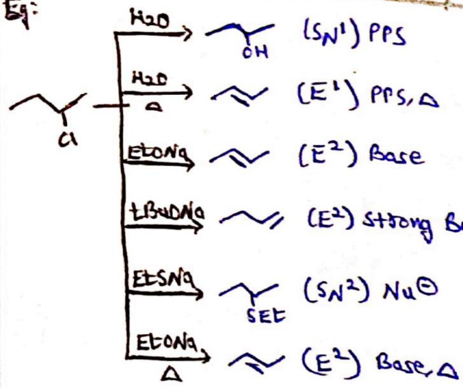
Substitution v/s Elimination:

S_N1 & E^1 → Carbocation intermediate and rearrangement possible

S_N2 & E^2 → One step Reactions & takes place only when strong Nu⊖/B⊖ (Base)

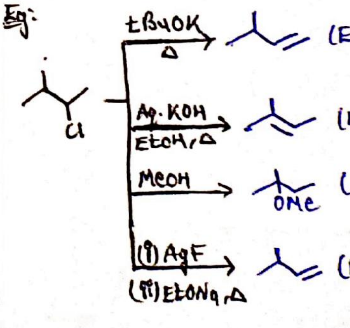
PPS	PPS/Δ	B⊖ (RO⊖, Alc-KOH)	B⊖, Δ	Nu⊖ (Aq-KOH)
↓	↓	↓	↓	↓
✓S_N1	✓E^1	✓E^2 (2°, 1°) ✓S_N2 (1° and unbranched)	✓E^2 ✓E-CB when EWG present	✓S_N2 (1°, 2°) ✓E^2 (3°) ✓S_N1 (3° + PPS)

Eg:



(S_N1) PPS
(E^1) PPS, Δ
(E^2) Base
(E^2) Strong Base
(S_N2) Nu⊖
(E^2) Base, Δ

Eg:



(E^2) Strong base, Δ
Alc. KOH
(E^2) Base, Δ
(SN1) PPS
(E^2) -F tha isly less stable alkene is major